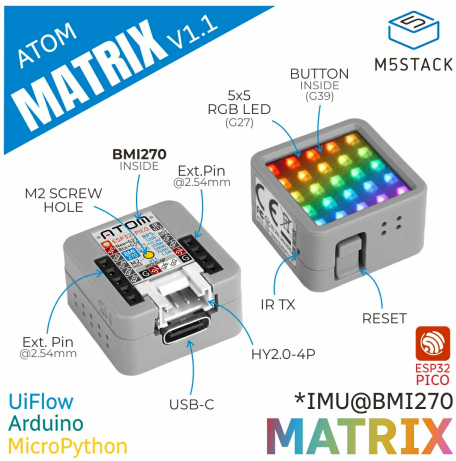
Atom-Matrix v1.1

SKU:C008-B-V11



























Description

Atom-Matrix v1.1 is an iterative upgrade of the Atom-Matrix. This product replaces the MPU6886 from the previous generation with a BMI270 six-axis IMU sensor, effectively improving attitude detection accuracy and response speed, making it better suited for high-precision applications such as motion capture and attitude control. The main controller adopts the ESP32-PICO-D4 solution with integrated Wi-Fi connectivity and onboard 4 MB SPI flash. It also features an infrared emitter and a 5×5 RGB LED matrix display. A programmable button is located beneath the LED matrix, supporting custom trigger logic for convenient implementation of IR remote control, visual interaction, and command triggering. This product is ideal for embedded development and prototyping scenarios that require high-precision attitude detection, infrared remote control, and visual interactive feedback.

Note

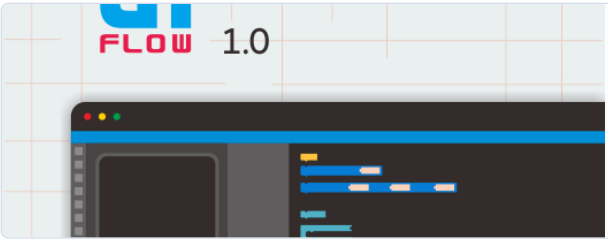
When using the RGB LEDs, it is recommended to set them to a moderate brightness level. Avoid setting excessively high brightness values, as this may damage the LEDs and the acrylic display.

Tutorial



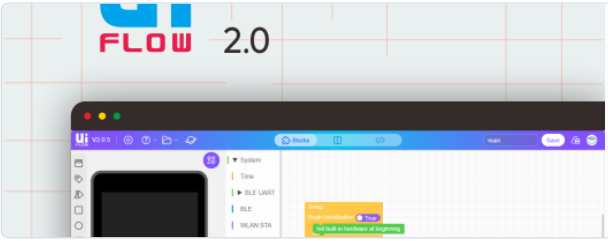
Arduino IDE

This tutorial introduces how to program and control the Atom-Matrix v1.1 using Arduino IDE.



UiFlow1

This tutorial introduces how to program and control the Atom-Matrix v1.1 using the UiFlow1 graphical programming platform.



UiFlow2

This tutorial introduces how to program and control the Atom-Matrix v1.1 using the UiFlow2 graphical programming platform.

Features

- Based on ESP32
- Compact form factor
- Built-in 3-axis gyroscope and 3-axis accelerometer (I2C Address: 0x68)
- Programmable button
- RGB LED matrix display
- Infrared transmission capability
- Expandable pins and interfaces
- Development Platform
  - UiFlow1
  - UiFlow2
  - Arduino IDE
  - ESP-IDF
  - PlatformIO

Includes

- 1 x Atom-Matrix v1.1

Applications

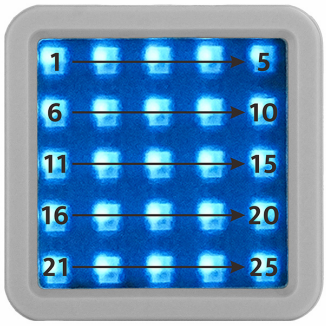
- Motion-sensing remote controller
- Motion capture / Attitude analysis
- Smart home central control node

Specifications

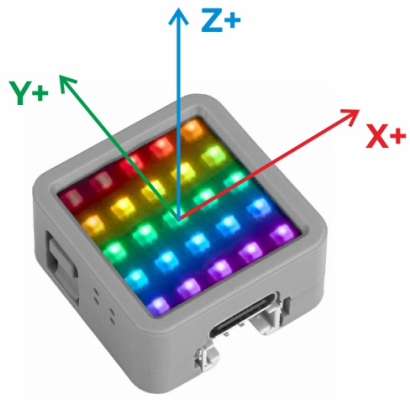
Specification	Parameter
SoC	ESP32-PICO-D4 @ Xtensa® 32-bit LX6 dual-core processor, clock frequency up to 240MHz
DMIPS	600
SRAM	520KB
Flash	4MB
Wi-Fi	2.4 GHz Wi-Fi
Input Voltage	5V @ 500mA
Host Interface	USB Type-C x 1, Grove (I2C+I/O+UART) x 1
RGB LED	25 x WS2812C-2020
MEMS	BMI270 (I2C Address: 0x68)
Infrared	IR transmission supported
Button	Programmable button x 1
Antenna	2.4G 3D Antenna
Operating Temp.	0 ~ 60°C
Power Consumption	5V @ 61.65mA
Product Size	24.0 x 24.0 x 13.9mm
Product Weight	7.3g
Package Size	85.0 x 66.0 x 15.0mm
Gross Weight	13.5g

Learn

RGB LED Matrix Pixel Order Diagram



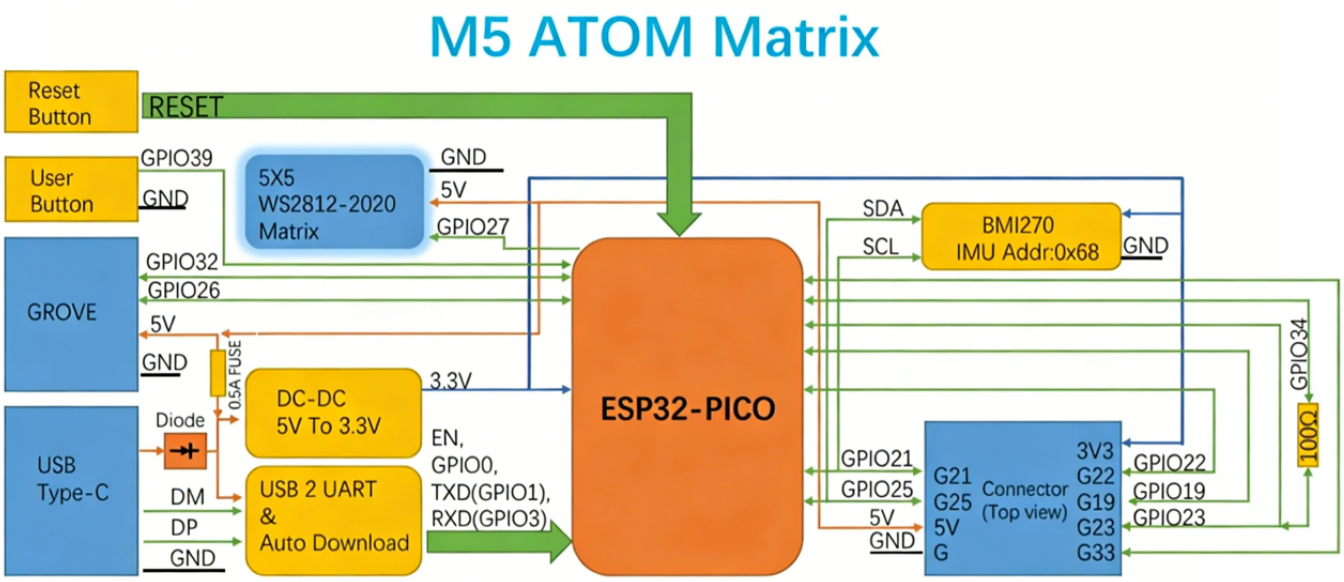
IMU Triaxial Direction Schematic Diagram

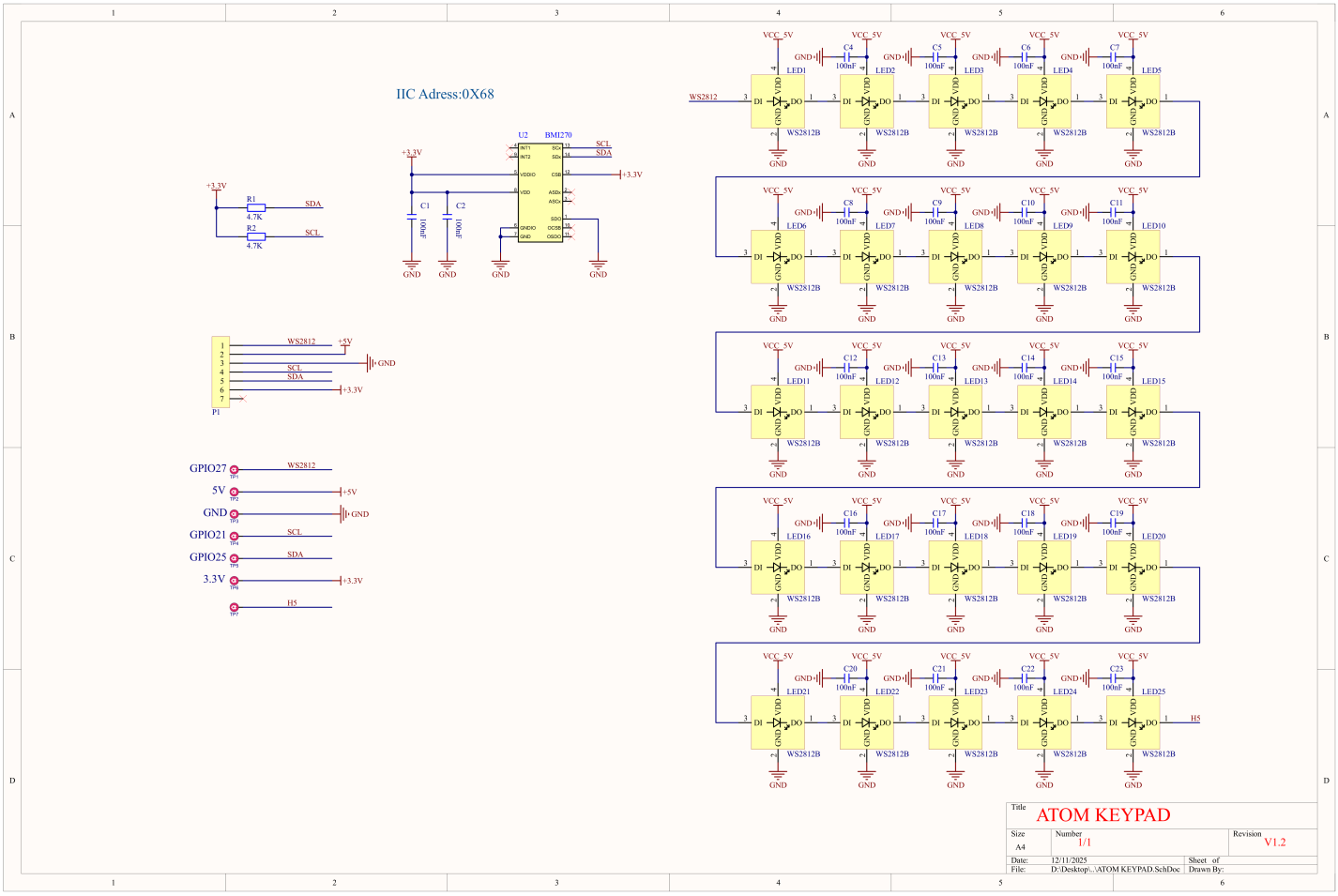


IMU ORIENTATION OF AXES

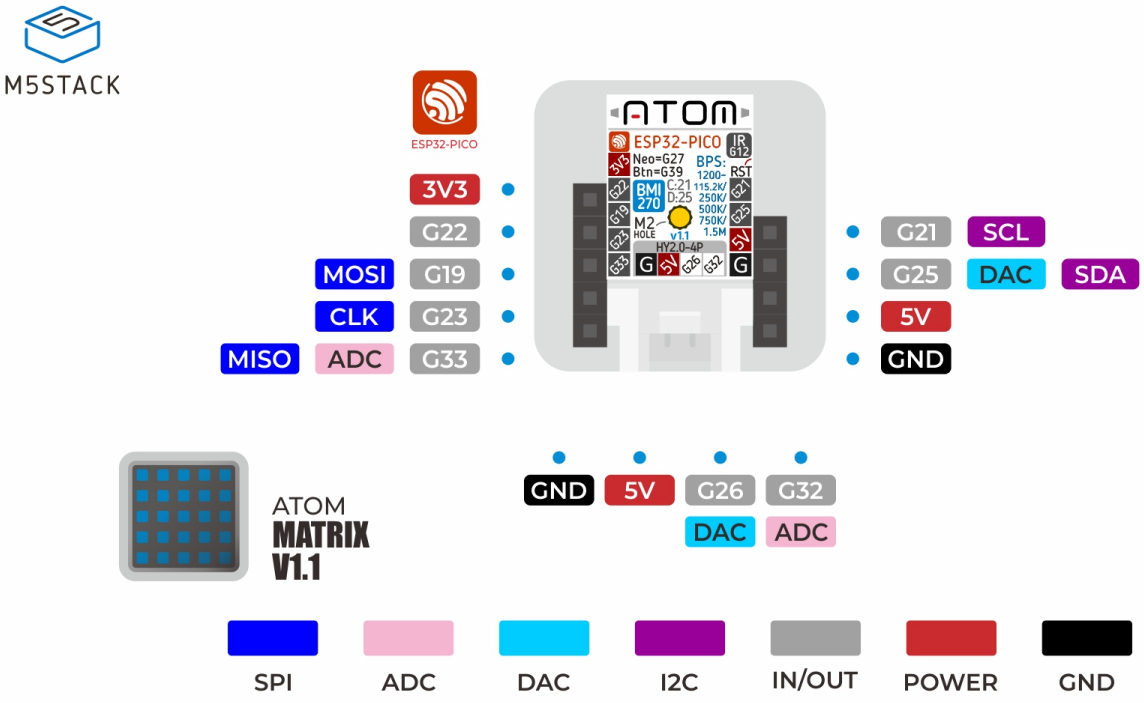
Schematics

- Atom-Matrix v1.1 LED Board Schematics PDF





PinMap



RGB & BUTTON & IR & BMI270

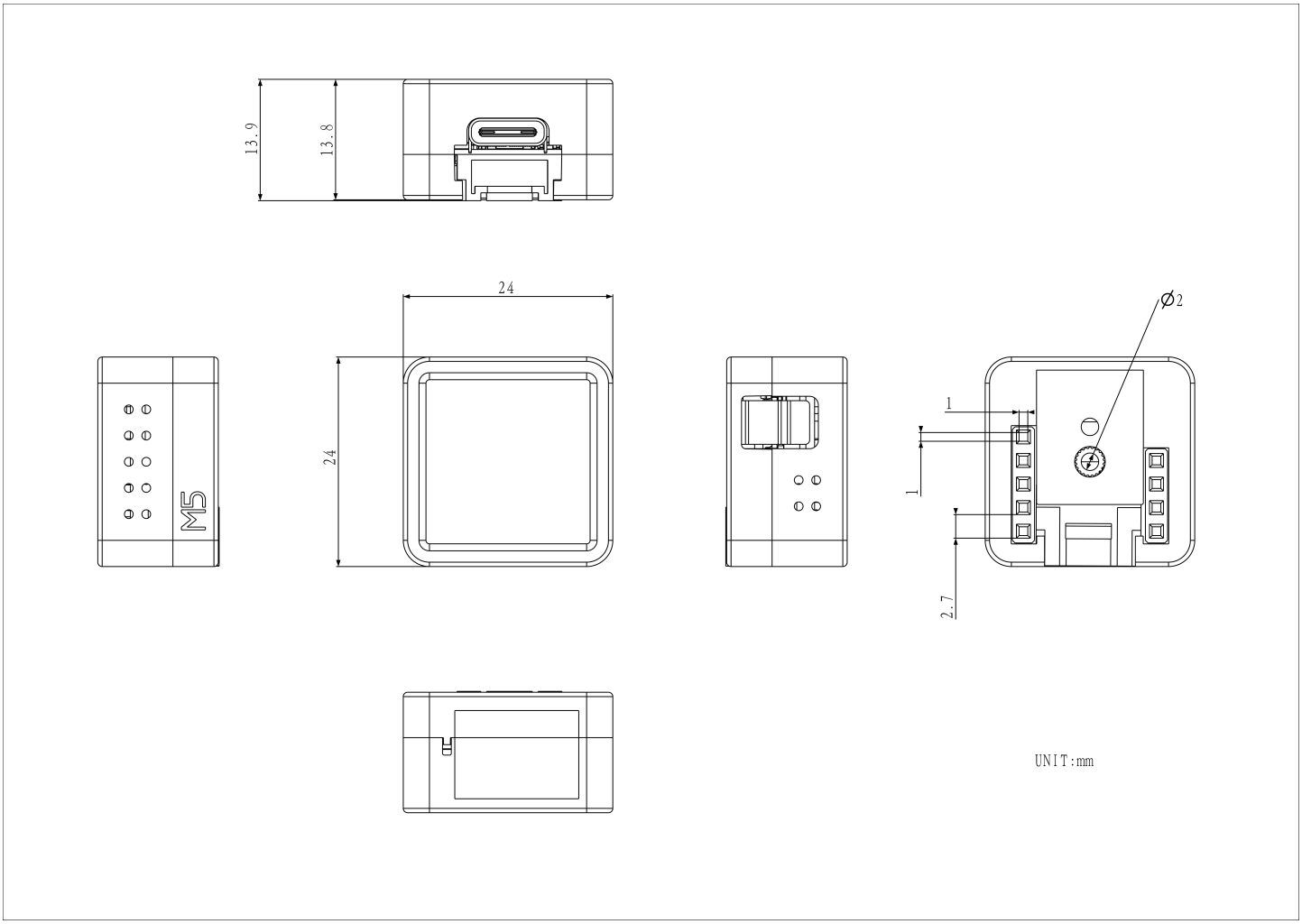
ESP32-PICO-D4	G27	G39	G12	G21	G25
RGB Led	Data				
Btn		Button			
IR			IR_TX		
BMI270 (0x68)				SCL	SDA

HY2.0-4P

HY2.0-4P	Black	Red	Yellow	White
PORT.CUSTOM	GND	5V	G26	G32

Model Size

- Atom-Matrix v1.1 Model Size PDF



Structure

- Atom-Matrix v1.1 Structure Files

Datasheets

- ESP32-PICO
- BMI270
- WS2812C-2020

Softwares

Arduino

- Atom-Matrix v1.1 Arduino Quick Start

UiFlow1

- Atom-Matrix v1.1 UiFlow1 Quick Start

UiFlow2

- Atom-Matrix v1.1 UiFlow2 Quick Start

PlatformIO

```
[env:m5stack-atom]
platform = espressif32@6.7.0
board = m5stack-atom
framework = arduino
upload_speed = 1500000
monitor_speed = 115200
build_flags =
    -DCORE_DEBUG_LEVEL=5
lib_deps =
    M5Unified=https://github.com/m5stack/M5Unified
```

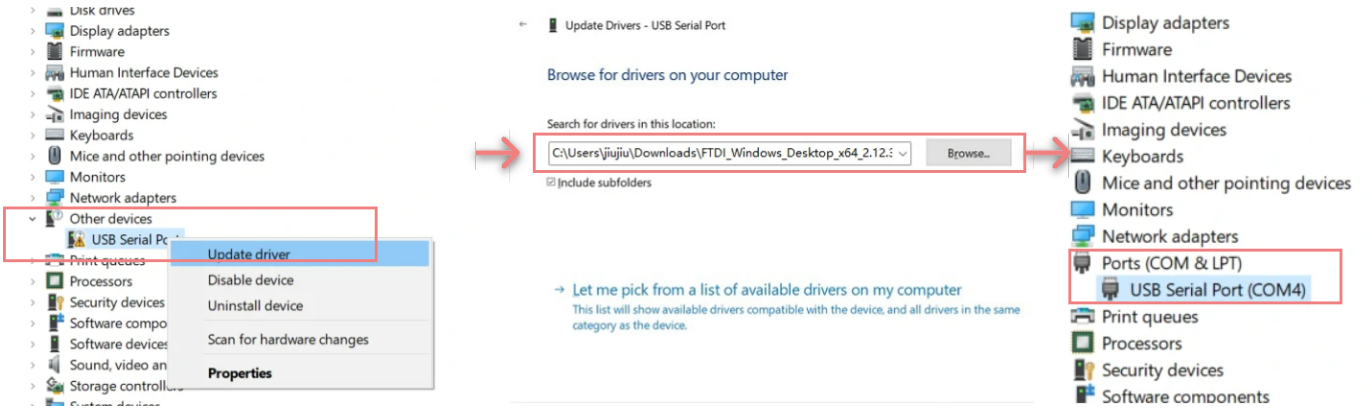
USB Driver

Atom-Matrix v1.1 may not work without a driver on some systems. You can resolve this issue by manually installing the FTDI VCP driver.

1. Select the driver file that matches your operating system from the list below, download it, and extract it. You can also visit the FTDI Chip official website for additional FTDI driver versions.

Driver Name	Download Link
Windows_Desktop_x64	Download
Windows_Desktop_x86_32	Download
Windows_Desktop_ARM	Download
MacOS_Intel	Download
MacOS_ARM64	Download

2. Connect the device to your PC, open Device Manager, and install the FTDI driver (the screenshot below uses Windows 10 as an example). (Note: On some systems, you may need to install the driver twice for it to take effect. The unrecognized device is usually named `M5Stack` or `USB Serial`.)



Video

Product Comparison

Product Compare	 Atom-Matrix v1.1	 Atom-Matrix
6-Axis IMU Sensor	BMI270	MPU6886

For a comparison of Atom series products, visit the Product Selector, check the target products to view the comparison results. The selector covers key information including core specifications and feature highlights, supporting simultaneous multi-product comparison.